Contracts to Effects Compiler Correctness

(Deep Handler)

Overview

Proof for compiler from Contracts to Untyped Deep Effects, the handler semantic is deep, where the continuation automatically reinstall the handler.

$\mathtt{h}_{\mathscr{C}} : \mathtt{lab} \quad = \quad \{\mathscr{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\}$

$\mathsf{wrap} \quad = \quad \mu w : \forall\alpha.\big(\big((\,) \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big) \to \alpha\big).\Lambda\alpha.\ \lambda f : \big((\,) \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big).$

$\mathtt{handler}\ \{\mathscr{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if } w[\mathsf{bool}]\ (\lambda_.e\ v)\ \text{then } k\ v\ \text{else } \mathtt{error}_l\}\ f$

where $\lambda\langle l, \langle e, v\rangle\rangle$ deconstructs the argument as tuple pattern

The proof get into the way a lot because it switch back and forth between states prepended with handle and states without handle.

Relations

Before we define relations that will be used in the simulation proof below. We define:

- Context E^* is any context without $\mathsf{check}^k_j(e, v)$ in Contracts.
- Context $E^+ = \Box \mid E^+\left[E^*\left[\mathsf{check}^k_j(\Box, v)\right]\right]$

$\boxed{R_v \subseteq \text{Contracts } v \times \mathtt{Effects}\ v}$

Abuse of notation: If $(v, v') \in R_v$, we write $v \sim_{R_v} v'$ or $v \sim v'$. We write $\tilde{v}$ for any value $\tilde{v}$ satisfying $v \sim \tilde{v}$.

$$
\begin{array}{rcl}
x & \sim & x \\
b & \sim & b \\
N & \sim & N \\
\lambda x : \tau.e & \sim & \lambda x : \tau^*.\tilde{e} \\
\mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v) & \sim & \tilde{e} \\
 & & e = \lambda x : \tau.\mathsf{mon}^{k,l}_j\left(\kappa_2, v\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\right) \\
 & & \text{With } \cdot \vdash \tau : v \\
\mathsf{guard}^{k,l}_j(\kappa_1 \to_i \lambda y.\kappa_2, v) & \sim & \tilde{e}
\end{array}
$$

$$e = \lambda x : \tau.\mathsf{mon}_j^{k,l}\Big(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big)$$
$$\kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big]$$
$$\text{With } \cdot \vdash \tau : v$$

$\boxed{R_e \subseteq \mathsf{Contracts}\ e \times \texttt{Effects}\ e}$

Abuse of notation: If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We write $\tilde{e}$ for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

$$\begin{aligned}
v &\sim \tilde{v} \\
e_1\ e_2 &\sim \tilde{e}_1\ \tilde{e}_2 \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) &\sim \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e) &\sim \tilde{e}_1\ \tilde{e} \\
&\quad e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e) &\sim \tilde{e}_1\ \tilde{e} \\
&\quad e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
&\quad \kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big] \\
\mathsf{blame}_j^k &\sim \texttt{error}_k
\end{aligned}$$

$\boxed{R_{E^*} \subseteq \mathsf{Contracts}\ E^* \times \texttt{Effects}\ E^*}$

Abuse of notation: If $(E^*, E'^*) \in R_{E^*}$, we write $E^* \sim_{R_{E^*}} E'^*$ or $E^* \sim E'^*$. We write $\tilde{E}^*$ for any expression $\tilde{E}^*$ satisfying $E^* \sim \tilde{E}^*$.

$$\begin{aligned}
\square &\sim \square \\
E^*\ e &\sim \tilde{E}^*\ \tilde{e} \\
v\ E^* &\sim \tilde{v}\ \tilde{E}^* \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E^*) &\sim \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{E}^* \rangle\rangle \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E^*) &\sim \tilde{e}_1\ \tilde{E}^* \\
&\quad e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E^*) &\sim \tilde{e}_1\ \tilde{E}^* \\
&\quad e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
&\quad \kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big]
\end{aligned}$$

$R_{E^+} \subseteq \mathsf{Contracts}\ E^+ \times \mathtt{Effects}\ E^+$

Abuse of notation: If $(E^+, E'^+) \in R_{E^+}$, we write $E^+ \sim_{R_{E^+}} E'^+$ or $E^+ \sim E'^+$. We write $\tilde{E}^+$ for any expression $\tilde{E}^+$ satisfying $E^+ \sim \tilde{E}^+$.

$$
\begin{array}{rcl}
\square & \sim & \square \\
E^+\left[E^*\left[\mathsf{check}^k_j(\square, v)\right]\right] & \sim & \tilde{E}^+\left[\text{if } \square \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \mathtt{error}_j\right] \\
E^+\, e & \sim & \tilde{E}^+\, e \\
v\, E^+ & \sim & v\, \tilde{E}^+ \\
\mathsf{mon}^{k,l}_j(\kappa, E^+) & \sim & \mathsf{mon}^{k,l}_j(\kappa, \tilde{E}^+)
\end{array}
$$

Lemma 0.1. (Relation is closed) If $e \sim e'$ and $E^* \sim E'^*$ then $E^*[e] \sim E'^*[e']$.

Proof. By induction on the structure of $E^* \sim E'^*$.

- Case $\square \sim \square$

  $E^*[e] = e$ — Context Plug in
  $E'^*[e'] = e'$ — Context Plug in
  $e \sim e'$ — Given
  $E^*[e] \sim E'^*[e']$

- Case $E^*_1\, e_1 \sim \tilde{E}^*_1\, \tilde{e}_1$

  $E^*_1 \sim \tilde{E}^*_1$ — Given
  $E^*_1[e] \sim \tilde{E}^*_1[e']$ — I.H.
  $E^*_1[e]\, e_1 \sim \tilde{E}^*_1[e']\, \tilde{e}_1$ — Relation
  $E^*[e] \sim E'^*[e']$

- Case $\mathsf{mon}^{k,l}_j(\mathsf{flat}(e_1), E^*_1) \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}^*_1\rangle\rangle$

  $E^*_1 \sim \tilde{E}^*_1$ — Given
  $E^*_1[e] \sim \tilde{E}^*_1[e']$ — I.H.
  $\mathsf{mon}^{k,l}_j(\mathsf{flat}(e_1), E^*_1[e]) \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}^*_1[e']\rangle\rangle$ — Relation
  $E^*[e] \sim E'^*[e']$

- Case $\mathsf{mon}^{k,l}_j\left(\kappa_1 \to \kappa_2, \tilde{E}_1\right) \sim \tilde{e}_1\, \tilde{E}^*_1$, with $e_1 = \lambda f.\lambda x.\mathsf{mon}^{k,l}_j\left(\kappa_2, f\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\right)$

  $e_1 \sim \tilde{e}_1$ — Given
  $E^*_1 \sim \tilde{E}^*_1$ — Given
  $E^*_1[e] \sim \tilde{E}^*_1[e']$ — I.H.
  $\mathsf{mon}^{k,l}_j(\kappa_1 \to \kappa_2, E^*_1[e]) \sim \tilde{e}_1\, \tilde{E}^*_1[e']$ — Relation

$E^*[e] \sim E'^*[e']$

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, \tilde{E}_1) \sim \tilde{e}_1\ \tilde{E}_1^*$, with $e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$

  Similarly to the above case

- Other cases follow similarly.

□

Lemma 0.2. (Relation decomposition through check-free contexts)

If $E^*[e] \sim e'$, then there exist E'^* and e'_0 such that $E^* \sim E'^*$ and $e \sim e'_0$ and $e' = E'^*[e'_0]$.

Proof. By induction on the check-free context E^*.

- Case $E^* = \square$

$E^*[e] = e \sim e'$	Given
$E'^* = \square$	Choose
$e'_0 = e'$	Choose
$E^* \sim E'^*$	Definition
$e \sim e'_0$	Given
$e' = E'^*[e'_0]$	Context Plug in

- Case $E^* = E_0^*\ e_1$

$E_0^*[e]\ e_1 \sim e'$	Given
$E_0^*[e] \sim e'_0$	Relation inversion
$e_1 \sim e'_1$	Relation inversion
$e' = e'_0\ e'_1$	Relation inversion
$E_0^* \sim E_0'^*$	I.H.
$e \sim e'_2$	I.H.
$e'_0 = E_0'^*[e'_2]$	I.H.
$E'^* = E_0'^*\ e'_1$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e'_2]$	Context Plug in

- Case $E^* = v\ E_0^*$

$v\ E_0^*[e] \sim e'$	Given
$v \sim v'$	Relation inversion
$E_0^*[e] \sim e'_0$	Relation inversion
$e' = v'\ e'_0$	Relation inversion

$E_0^* \sim E_0'^*$	I.H.
$e \sim e_1'$	I.H.
$e_0' = E_0'^*[e_1']$	I.H.
$E'^* = v'\ E_0'^*$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e_1']$	Context Plug in

- Case $E^* = \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_0), E_0^*)$

$\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_0), E_0^*[e]) \sim e'$	Given
$e_0 \sim e_0'$	Relation inversion
$E_0^*[e] \sim e_1'$	Relation inversion
$e' = \mathtt{perform}\ \mathscr{C}\ \langle k, \langle e_0', e_1' \rangle \rangle$	Relation inversion
$E_0^* \sim E_0'^*$	I.H.
$e \sim e_2'$	I.H.
$e_1' = E_0'^*[e_2']$	I.H.
$E'^* = \mathtt{perform}\ \mathscr{C}\ \langle k, \langle e_0', E_0'^* \rangle \rangle$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e_2']$	Context Plug in

- Case $E^* = \mathsf{mon}_j^{k,l}(\kappa, E_0^*)$

$\mathsf{mon}_j^{k,l}(\kappa, E_0^*[e]) \sim e'$	Given
$E_0^*[e] \sim e_0'$	Relation inversion
$e' = e_c'\ e_0'$	Relation inversion
$E_0^* \sim E_0'^*$	I.H.
$e \sim e_1'$	I.H.
$e_0' = E_0'^*[e_1']$	I.H.
$E'^* = e_c'\ E_0'^*$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e_1']$	Context Plug in

- Other cases follow similarly.

□

Lemma 0.3. (Relation is closed under substitution)

If $e \sim e'$ and $v \sim v'$ then $e[x := v] \sim e'[x := v']$.

Proof. By induction on the derivation of $e \sim e'$.

- Case $x \sim x$

$x[x := v] = v$ Replacement
$x[x := v'] = v'$ Replacement
$v \sim v'$ Given
$x[x := v] \sim x[x := v']$

- Case $z \sim z$, where $z \neq x$

$z[x := v] = z$ Replacement
$z[x := v'] = z$ Replacement
$z[x := v] \sim z[x := v']$ Definition

- Case $\lambda z.e_1 \sim \lambda z.e_1'$

$e_1[x := v] \sim e_1'[x := v']$ I.H.
$\lambda z.e_1[x := v] \sim \lambda z.e_1'[x := v']$ Definition
$(\lambda z.e_1)[x := v] \sim (\lambda z.e_1')[x := v']$ Replacement

- Case $e_1\ e_2 \sim e_1'\ e_2'$

$e_1[x := v] \sim e_1'[x := v']$ I.H.
$e_2[x := v] \sim e_2'[x := v']$ I.H.
$e_1\ e_2[x := v] = e_1[x := v]\ e_2[x := v]$ Replacement
$e_1'\ e_2'[x := v'] = e_1'[x := v']\ e_2'[x := v']$ Replacement
$e_1\ e_2[x := v] \sim e_1'\ e_2'[x := v']$ Definition

- Case $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) \sim \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle \rangle$

$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
$e_2[x := v] \sim \tilde{e}_2[x := v']$ I.H.
$\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2)[x := v] \sim \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle \rangle[x := v']$ Definition

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\right)$

$e_1 \sim \tilde{e}_1$ Given
$e_2 \sim \tilde{e}_2$ Given
$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
$e_2[x := v] \sim \tilde{e}_2[x := v']$ I.H.
$\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] = \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to \kappa_2[x := v], e_1[x := v])$
$e_2[x := v] = \lambda f.\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
$\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ Relation
$\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$

Choose variables, $f, y \neq x$ and $f, y \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, \tilde{e}_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$ and $y \neq x$

  $e_1 \sim \tilde{e}_1$ — Given
  $e_2 \sim \tilde{e}_2$ — Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ — I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] =$ — Substitution

  $\quad \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to_i \lambda y.\kappa_2[x := v], e_1[x := v])$

  $e_2[x := v] = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], z)\right)$
  $\kappa_3[x := v] = \left(\kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]\right)[x := v]$
  $\kappa_3[x := v] = (\kappa_2[x := v])\left[y := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$ — Substitution?
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ — Relation
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$ — Definition

  Choose variables, $f, z \neq x$ and $f, y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, \tilde{e}_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$

  $e_1 \sim \tilde{e}_1$ — Given
  $e_2 \sim \tilde{e}_2$ — Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ — I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] =$ — Substitution

  $\quad \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to_i \lambda x.\kappa_2, e_1[x := v])$

  $e_2[x := v] = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
  $\kappa_3[x := v] = \left(\kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]\right)[x := v]$
  $\kappa_3[x := v] = \left(\kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]\right)$
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ — Relation
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$ — Definition

  Choose variables, $f, z \neq x$ and $f, y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1) \sim \tilde{e}_1$, with $e_1 = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\right)$

  $e_1 \sim \tilde{e}_1$ — Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
  $\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)[x := v] = \mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to \kappa_2[x := v], v_1[x := v])$
  $\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)[x := v] \sim \tilde{e}_1[x := v']$ — Relation

Choose variables, $y \neq x$ and $y \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \sim \tilde{e}_1$, with $e_1 = \lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$

  $e_1 \sim \tilde{e}_1$ Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
  $e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], x)\right)$
  $\kappa_3[x := v] = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right][x := v]$
  $\kappa_3[x := v] = (\kappa_2[x := v])\left[y := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$ Substitution?
  $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)[x := v] =$

  $\mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to_i \lambda y.\kappa_2[x := v], v_1[x := v])$

  $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)[x := v] \sim \tilde{e}[x := v']$ Relation

  Choose variables, $z \neq x$ and $y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v) \sim \tilde{e}_1$, with $e_1 = \lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ and $\kappa_3 = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$

  $e_1 \sim \tilde{e}_1$ Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
  $e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], x)\right)$
  $\kappa_3[x := v] = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right][x := v]$
  $\kappa_3[x := v] = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$
  $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)[x := v] =$

  $\mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to_i \lambda x.\kappa_2, v_1[x := v])$

  $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)[x := v] \sim \tilde{e}[x := v']$ Relation

  Choose variables, $z \neq x$ and $z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Other cases follows similarly.

□

Lemma 0.4. (Breaking down Context E)

For all contexts E, uniquely decompose $E = E^+[E^*]$.

Uniqueness observation: Recall that E^* has no check contexts.

- If E does not have check, then it is $E^* = E$, and $E^+ = \square$.
- Else there is only a single decomposition to get out E^* without check contexts

Proof. By induction on E.

- Case $E = \square$

Choose $E^+ = E^* = \square$

- Case $E = E_1\ e$

  $E_1 = E_1^+[E_1^*]$ I.H.
  $E = E_1^+[E_1^*]\ e$
  $E = (E_1^+\ e)[E_1^*]$

  - Case $E_1^+ = \square$

    $E = E_1^*\ e$
    $E^+ = \square$ Choose
    $E^* = E_1^*\ e$ Choose

  - Case $E_1^+ \neq \square$

    $E^+ = E_1^+\ e$ Lemma 0.5 ($K = \square\ e$) and Choose
    $E^* = E_1^*$ Choose

- Case $E = \mathsf{check}_j^k(E_1, v)$

  $E_1 = E_1^+[E_1^*]$ I.H.
  $E = \mathsf{check}_j^k(E_1^+[E_1^*], v) = \mathsf{check}_j^k(E_1^+, v)[E_1^*]$

  - Case $E_1^+ = \square$

    $E = \mathsf{check}_j^k(\square, v)[E_1^*]$
    $E^+ = \mathsf{check}_j^k(\square, v), E^* = E_1^*$ Choose

  - Case $E_1^+ \neq \square$

    $E_2^+ = \mathsf{check}_j^k(E_1^+, v)$ Lemma 0.5 ($K = \mathsf{check}_j^k(\square, v)$)
    $E = E_2^+[E^*]$

□

Lemma 0.5.

If $E_1^+ \neq \square$ then there is a E_2^+, $E_2^+ = K[E_1^+]$.

Where $K = \square\ e \mid v\ \square \mid \mathsf{mon}_j^{k,l}(\kappa, \square) \mid \mathsf{check}_j^k(\square, v)$

Proof.

By induction on the structure of E_1^+.

- Case $\square$ invalid.
- Case $E_3^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$

  - Case $E_3^+ = \square$

    – Case $K = \mathsf{check}_{j_1}^{k_1}(\square, v_1)$

$K = E_4^+$ Definition of E^+

$K[E_1^+] = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$

$E_2^+ = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$ Choose

– Case $K \neq \mathsf{check}_{j_1}^{k_1}(\square, v_1)$

$K = E_4^*$ Definition of E^*

$K[E_1^+] = E_4^*[E^*[\mathsf{check}_j^k(\square, v)]] = E_5^*[\mathsf{check}_j^k(\square, v)]$

$E_2^+ = \square[E_5^*[\mathsf{check}_j^k(\square, v)]]$ Choose

▸ Case $E_3^+ \neq \square$

$E_4^+ = K[E_3^+]$ I.H.

$K[E_1^+] = K[E_3^+[E^*[\mathsf{check}_j^k(\square, v)]]] =$

$K[E_3^+][E^*[\mathsf{check}_j^k(\square, v)]] = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$

$E_2^+ = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$ Choose

□

Lemma 0.6. (Compiler is subset of relation)

If $\Gamma \vdash e_1 : \tau \rightarrowtail e_2$ then $e_1 \sim e_2^*$.

In particular, for closed programs, if $\cdot \vdash e_1 : \tau \rightarrowtail e_2$ then $e_1 \sim e_2^*$.

In this lemma, we use e^* as shorthand to denote e untyped.

Proof.

By induction on structure of $\Gamma \vdash e_1 : \tau \rightarrowtail e_2$.

- Case

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau \rightarrowtail x}$$

$e_2^* = x$ Definition

$x \sim x$ Relation

- Case

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \rightarrowtail e'}{\Gamma \vdash \lambda x : \tau_1.e : \tau_1 \to \tau_2 \rightarrowtail \lambda x : \tau_1.e'}$$

$e_2^* = \lambda x.e'^*$ Definition

$e \sim e'^*$ I.H. on the premise with context $\Gamma, x : \tau_1$

$\lambda x.e \sim \lambda x.e'^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau \to \mathsf{bool} \rightarrowtail e_5 \qquad \Gamma \vdash e_4 : \tau \rightarrowtail e_6}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_3), e_4) : \tau \rightarrowtail \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_5, e_6\rangle\rangle}$$

$e_2^* = \mathtt{perform}\ \mathscr{C}\ \langle k, \langle e_5^*, e_6^*\rangle\rangle$ Definition
$e_3 \sim e_5^*$ I.H.
$e_4 \sim e_6^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau_2 \rightarrowtail e_4 \qquad \Gamma \vdash \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_5\ e_4}$$

$e_3 \sim e_4^*$ I.H.
$\lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \sim e_5^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau_1 \to \tau_2 \rightarrowtail e_4 \qquad \Gamma \vdash \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_3) : \tau_1 \to \tau_2 \rightarrowtail e_5\ e_4}$$

$e_3 \sim e_4^*$ I.H.
$\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \sim e_5^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Other cases are similar.

□

Compiler Correctness

Figure 1 shows the structure of the proof. The source program e compiles to e', and the target program runs e' under wrap. After the handler is installed, the target reaches a state e_0'' with $\mathrm{EQ}(e, e_0'')$ (Lemma 0.8). From then on, every source step $e_i \to e_{i+1}$ is matched by zero or more target steps $e_i'' \Longrightarrow^* e_{i+1}''$ that keep $\mathrm{EQ}(e_{i+1}, e_{i+1}'')$ (Lemma 0.10). Chaining the squares (Lemma 0.9), the source result r and the target result $\tilde{r}$ are related: a value goes to its related value, and blame_j^k goes to $\mathtt{error}_k$.

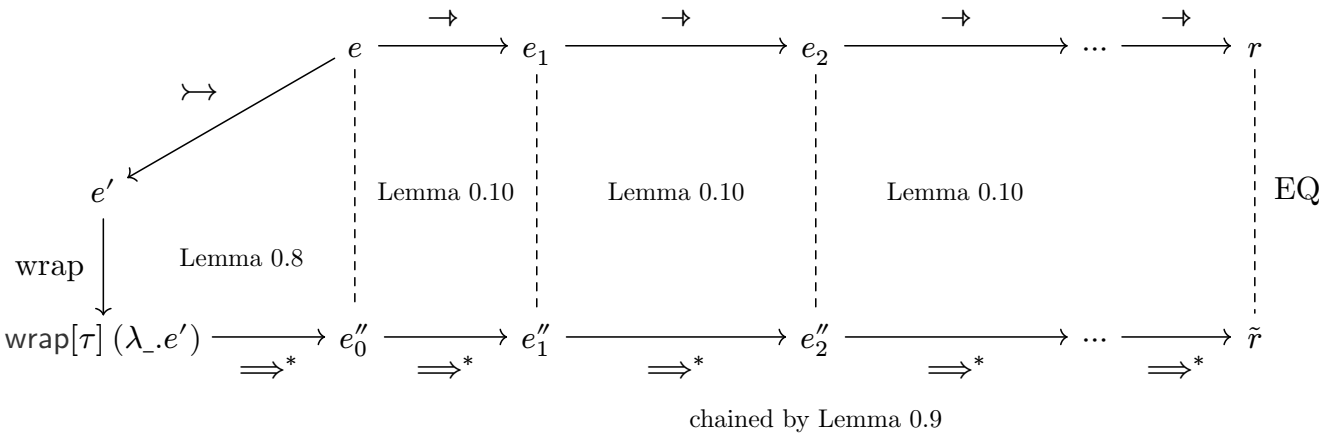


Figure 1: Structure of the compiler correctness proof. Dashed lines rel states by EQ.

Define a predicate between Contracts e and Untyped Effects e'.

$$\mathrm{EQ}(e, e') = \begin{cases} \text{true if } e = v \text{ and } e' = \tilde{v} \\ \text{true if } e' = \texttt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{e} \text{ and } e \neq E^*\left[\mathsf{check}^k_j(e'', v)\right] \\ \text{true if } e = E^+\left[E^*\left[\mathsf{check}^k_j(e_1, v)\right]\right] \text{ and} \\ \qquad e' = \tilde{E}^+\left[\text{if } e'_1 \text{ then } \left(\lambda x.\ \texttt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \texttt{error}_j\right] \text{ and} \\ \qquad \mathrm{EQ}(e_1, e'_1) \\ \text{true if } e = \mathsf{blame}^k_j \text{ and } e' = \texttt{error}_k \\ \text{false otherwise} \end{cases}$$

Theorem 0.7. (Compiler Correctness)

If $\cdot \vdash e : \tau \rightarrowtail e'$ and $e^* = \mathrm{untyped}(\mathsf{wrap}[\tau]\ (\lambda_.e'))$ and $e \twoheadrightarrow^* b$ then $e^* \Longrightarrow^* b$.

Proof.

$e^* \Longrightarrow^* e'_1$	Lemma 0.8
$\mathrm{EQ}(e, e'_1)$	Lemma 0.8
$e'_1 \Longrightarrow^* b$	Lemma 0.9
$e^* \Longrightarrow^* b$	Combining steps

□

Lemma 0.8. (Wrapped program steps into an EQ state with source)

If $\cdot \vdash e : \tau \rightarrowtail e'$ and $e_1^* = \mathrm{untyped}(\mathsf{wrap}[\tau]\ (\lambda_.e'))$ then it exists e_2^* such that $e_1^* \Longrightarrow^* e_2^*$ and $\mathrm{EQ}(e, e_2^*)$.

In this lemma, we use e^* as shorthand to denote e untyped.

Proof.

$e \sim e'^*$	Lemma 0.6

$e_1^* \Longrightarrow^* \texttt{handler}\ \texttt{h}_{\mathscr{C}}\ \lambda_.e'^* \Longrightarrow^* \texttt{handle}\ \texttt{h}_{\mathscr{C}}\ e'^* = e_2^*$ — `Effects` reduction
$\mathrm{EQ}(e, e_2^*)$ — Definition

□

Lemma 0.9. If $e \twoheadrightarrow^* r$ and $\mathrm{EQ}(e, e')$ then $e' \Longrightarrow^* \tilde{r}$.

Proof. By breaking down the $\twoheadrightarrow^*$, and induction on it.

- Case $v \twoheadrightarrow^* v$

  $e' = \tilde{r}$ — Definition of $\mathrm{EQ}(e, e')$
  $e' \Longrightarrow^* \tilde{r}$ — Reflexivity

- Case $\mathsf{blame}_j^k \twoheadrightarrow^* \mathsf{blame}_j^k$

  $r = \mathsf{blame}_j^k$ — Case assumption
  $\tilde{r} = \texttt{error}_j$ — Relation
  $e' = \texttt{error}_j = \tilde{r}$ — Definition of $\mathrm{EQ}(e, e')$
  $e' \Longrightarrow^* \tilde{r}$ — Reflexivity

- Case $e \twoheadrightarrow e_1$ and $e_1 \twoheadrightarrow^* r$

  $e' \Longrightarrow^* e_1'$ — Lemma 0.10
  $\mathrm{EQ}(e_1, e_1')$ — Lemma 0.10
  $e_1' \Longrightarrow^* \tilde{r}$ — I.H.
  $e' \Longrightarrow^* \tilde{r}$ — Chaining steps

□

We also need to make a claim that in no where in e_1' do we have $\texttt{perform}\ \texttt{eff}$ that $\texttt{eff} \neq \mathscr{C}$.

Lemma 0.10. If $e_1 \twoheadrightarrow e_2$ and $\mathrm{EQ}(e_1, e_1')$ then there exists e_2' such that $e_1' \Longrightarrow^* e_2'$ and $\mathrm{EQ}(e_2, e_2')$.

Proof of Lemma 0.10. By case analysis on $e_1 \twoheadrightarrow e_2$ with respect to $\mathrm{EQ}(e_1, e_1')$.

- Case $E\left[\mathsf{blame}_j^k\right] \twoheadrightarrow \mathsf{blame}_j^k$

  $E = E^+[E^*]$ — Lemma 0.4

  ▸ Case $E^+ = \square$

  $E^*\left[\mathsf{blame}_j^k\right] \sim \tilde{E}^*[\texttt{error}_k]$ — Lemma 0.2
  $\tilde{E}^*[\texttt{error}_k] \Longrightarrow \texttt{error}_k$
  $\mathsf{blame}_j^k \sim \texttt{error}_k$ — Definition
  $\mathrm{EQ}\left(\mathsf{blame}_j^k, \texttt{error}_k\right)$ — EQ Definition
  $e_2' = \texttt{error}_k$ — Choose

- Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

  $E\left[\mathsf{blame}_j^k\right] = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(\mathsf{blame}_j^k, v_1\right)\right]\right]$

  $\mathsf{blame}_j^k \sim \mathtt{error}_k$ Definition

  $e_1 \sim \tilde{E}_1^+\left[\text{if } \mathtt{error}_k \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}_1^*[x]\right)\tilde{v} \text{ else } \mathtt{error}_{k_1}\right]$

  $e_1' \Longrightarrow \mathtt{error}_k$

  $\mathrm{EQ}\left(\mathsf{blame}_j^k, \mathtt{error}_k\right)$ EQ Definition

  $e_2' = \mathtt{error}_k$ Choose

- Case $E[e_3] \to E[e_4]$ where $e_3 \dashrightarrow e_4$

  Overview of proof strategy: breaking down $E = E^+[E^*]$ using Lemma 0.4 and case on E^+, then try to match $\mathrm{EQ}(e_1, e_1')$ cases. EQ case 2 holds if and only if $E^+ = \square$, and case 3 holds if and only if $E^+ \neq \square$ (other case) easily observed the form of e_1 in EQ. By step it should arrive at the next state that matches EQ predicate cases.

  During the reduction, we likely encounter $\mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[e]$. By Lemma 0.11, the handler invoked is obviously the current handler.

  - Case $e_3 = \mathsf{mon}_j^{k,l}(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}_j^k(e\ v, v) = e_4$

    $E = E^+[E^*]$ Lemma 0.4

    $e_1 = E^+[E^*[e_3]]$ Substitution

    – Case $E^+ = \square$

      $E^* \sim \tilde{E}^*$ Lemma 0.2

      $e_3 \sim \tilde{e}_3$ Lemma 0.2

      $\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2

      $e_1 = E^*[e_3] \sim \tilde{E}^*[\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle]$ Relation

      $e_1' = \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle]$ Case 2

      $e_1' \Longrightarrow^* \text{if } \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \mathtt{error}_j$

      $e_2' = \text{if } \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \mathtt{error}_j$ Choose

      $\mathrm{EQ}(e\ v, \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v})$ Case 2

      $\mathrm{EQ}(e_2, e_2')$ Case 3

    – Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

      $e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(E^*[e_3], v_1)\right]\right]$

      $e_3 \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle = \tilde{e}_3$

      $\mathrm{EQ}\left(E^*[e_3], \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_3]\right)$ Case 2, by shape

      $\mathrm{EQ}\left(e_1, \tilde{E}_1^+\left[\text{if } \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_3] \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}_1^*[x]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]\right)$

      $e_1' = \tilde{E}_1^+\left[\text{if } \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_3] \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}_1^*[x]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]$

      Identified e_1' based on the condition of EQ case 3

$e_1' \Longrightarrow^* \tilde{E}_2^+\left[\text{if } \texttt{handle h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \texttt{error}_j\right]$
$e_2' = \tilde{E}_2^+\left[\text{if } \texttt{handle h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \texttt{error}_j\right]$ Choose
where $\tilde{E}_2^+ = \tilde{E}_1^+\left[\text{if } \square \text{ then } \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}_1^*[x]\right)\tilde{v}_1 \text{ else } \texttt{error}_{j_1}\right]$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{check}_j^k(e\ v, v)\right], v_1\right)\right]\right]$
$\mathrm{EQ}\left(E^*\left[\mathsf{check}_j^k(e\ v, v)\right], \text{if } \texttt{handle h}_{\mathscr{C}}\ \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v} \text{ else } \texttt{error}_j\right)$
$\mathrm{EQ}(e_2, e_2')$ Case 3

(Case 2 is invalid, because of the shape of e_2)

- Case $e_3 = \mathsf{check}_j^k(\mathsf{true}, v) \dashrightarrow v = e_4$

$E = E^+[E^*]$ Lemma 0.4
$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

Let $e_5 = \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v}$.

$e_1' = \text{if } e_3' \text{ then } e_5 \text{ else } \texttt{error}_j$ $\mathrm{EQ}(e_1, e_1')$, case 3
$\mathrm{EQ}(\mathsf{true}, e_3')$ Since true is a value
$e_3' = \mathsf{true}$ Definition of EQ
$e_1' \Longrightarrow^* \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{v}] = e_2'$
$e_2 = E^*[v]$
$\mathrm{EQ}(e_2, e_2')$ Case 2

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

Let $e_5 = \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\tilde{v}$ and $e_6 = \left(\lambda x.\ \texttt{handle h}_{\mathscr{C}}\ \tilde{E}_1^*[x]\right)\tilde{v}_1$.

$e_1' =$ $\mathrm{EQ}(e_1, e_1')$, case 3
$\tilde{E}_1^+\left[\text{if if } e_3' \text{ then } e_5 \text{ else } \texttt{error}_j \text{ then } e_6 \text{ else } \texttt{error}_{j_1}\right]$
$\mathrm{EQ}(\mathsf{true}, e_3')$ Since true is a value
$e_3' = \mathsf{true}$ Definition of EQ
$e_1' \Longrightarrow^* \tilde{E}_1^+\left[\text{if } \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{v}] \text{ then } e_6 \text{ else } \texttt{error}_{j_1}\right] = e_2'$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(E^*[v], v_1)\right]\right]$
$\mathrm{EQ}\left(E^*[v], \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{v}]\right)$ Case 2
$\mathrm{EQ}(e_2, e_2')$ Case 3

(Case 2 is invalid, because of the shape of e_2.)

- Case $e_3 = \mathsf{check}_j^k(\mathsf{false}, v) \dashrightarrow \mathsf{blame}_j^k = e_4$

$E = E^+[E^*]$ Lemma 0.4
$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{check}^k_j(\mathsf{false}, v)\right]$

$e'_1 =$

– Case $E^+ = E^+_1\left[E^*_1\left[\mathsf{check}^{k_1}_{j_1}(\square, v_1)\right]\right]$

$e_1 = E^+\left[E^*\left[\mathsf{check}^k_j(\mathsf{false}, v)\right]\right]$

$e'_1 = \tilde{E}^+\left[\text{if false then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[x]\right)\ \tilde{v} \text{ else } \mathtt{error}_j\right]$ EQ(e_1, e'_1)

$e'_1 \Longrightarrow^* \tilde{E}^+\left[\mathtt{error}_j\right] = e'_2$ Reduction

$e_2 = E^+\left[E^*\left[\mathsf{blame}^k_j\right]\right]$ Source reduction

$\tilde{E}^+\left[\mathtt{error}_j\right] \Longrightarrow \mathtt{error}_j$ `Step-E-Error`

$\mathsf{blame}^k_j \sim \mathtt{error}_j$ Relation

EQ(e_2, e'_2) Definition

▸ Case $e_3 = \mathsf{mon}^{k,l}_j(\kappa_1 \to \kappa_2, v) \dashrightarrow \mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v) = e_4$

$E = E^+[E^*]$ Lemma 0.4

$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

Let $e_5 = \lambda f.\lambda x.\mathsf{mon}^{k,l}_j\left(\kappa_2, f\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\right) = \lambda f.\lambda x.e_{5a}$

$E^* \sim \tilde{E}^*$ Lemma 0.2

$e_3 \sim \tilde{e}_3$ Lemma 0.2

$\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2

$e_1 = E^*[e_3] \sim \tilde{E}^*[(\lambda f.\lambda x.\tilde{e}_{5a})\ \tilde{v}]$ Relation

$e'_1 = \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[(\lambda f.\lambda x.\tilde{e}_{5a})\ \tilde{v}]$ EQ(e_1, e'_1), case 2

$e'_1 \Longrightarrow^* \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\lambda x.\tilde{e}_{5a}[f := \tilde{v}]] = e'_2$ Reduction

$e_2 = E^*\left[\mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v)\right]$

$e_6 = \lambda x.\mathsf{mon}^{k,l}_j\left(\kappa_2, v\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\right)$ Let

$e_6 = e_5[f := v]$ Substitution

$\tilde{e}_6 = \tilde{e}_5[f := \tilde{v}]$ Lemma 0.3

$\mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_6 = \tilde{e}_5[f := \tilde{v}]$ Definition

$E^*\left[\mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_5[f := \tilde{v}]]$ Lemma 0.1

EQ(e_2, e'_2) Case 2

– Case $E^+ = E^+_1\left[E^*_1\left[\mathsf{check}^{k_1}_{j_1}(\square, v_1)\right]\right]$

Let $e_5 = \lambda f.\lambda x.e_{5a}$, where $e_{5a} = \mathsf{mon}^{k,l}_j\left(\kappa_2, f\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\right)$.

Let $\tilde{E}^+_2 = \tilde{E}^+_1\left[\text{if } \square \text{ then } \left(\lambda x.\ \mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*_1[x]\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]$

$e'_1 = \tilde{E}^+_2\left[\mathtt{handle}\ \mathtt{h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_5\ \tilde{v}]\right]$

$e_1' \Longrightarrow^* \tilde{E}_2^+\left[\texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\lambda x.\tilde{e}_{5a}[f := \tilde{v}]]\right] = e_2'$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], v_1\right)\right]\right]$
$e_6 = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ Let
$e_6 = e_{5a}[f := v]$ Substitution
$\tilde{e}_6 = \tilde{e}_{5a}[f := \tilde{v}]$ Lemma 0.3
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_6 = \tilde{e}_{5a}[f := \tilde{v}]$ Definition
$E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_{5a}[f := \tilde{v}]]$ Lemma 0.1
$\mathrm{EQ}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_{5a}[f := \tilde{v}]]\right)$ Case 1
$\mathrm{EQ}(e_2, e_2')$ Case 3

- Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) = e_4$

  $E = E^+[E^*]$ Lemma 0.4
  $e_1 = E^+[E^*[e_3]]$ Substitution

  – Case $E^+ = \square$

    Let $e_5 = \lambda f.\lambda x.e_{5a}$, where $e_{5a} = \mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$.

    $E^* \sim \tilde{E}^*$ Lemma 0.2
    $e_3 \sim \tilde{e}_3$ Lemma 0.2
    $\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2
    $\tilde{e}_3 = \tilde{e}_5\ \tilde{v}_2$ Relation
    $e_1' = \texttt{handle h}_{\mathscr{C}}\ \tilde{E}^*[\tilde{e}_3]$ $\mathrm{EQ}(e_1, e_1')$, case 2
    $e_1' \Longrightarrow^* e_1' = e_2'$ Reflexivity
    $\tilde{e}_4 = \tilde{e}_5\ \tilde{v}_2$ Relation
    $E^*[e_4] \sim \tilde{E}^*[\tilde{e}_4]$ Lemma 0.1
    $\mathrm{EQ}(e_2, e_2')$ Case 2

  – Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

    Same as the case $E^+ = \square$, inside $\tilde{E}^+$, by case 3.

- Case $e_3 = \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) = e_4$

  Similar to its functional variant.

- Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) = e_4$ where $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\right]$

  Similar to its functional variant.

□

Lemma 0.11. ($\tilde{E}^*$ does not have a handler)

If $E^* \sim \tilde{E}^*$, then $\mathscr{C} \notin \mathsf{bop}(\tilde{E}^*)$.

Proof.

By induction on the structure of $E^* \sim \tilde{E}^*$. Trivial, the related context has no handler.$\square$